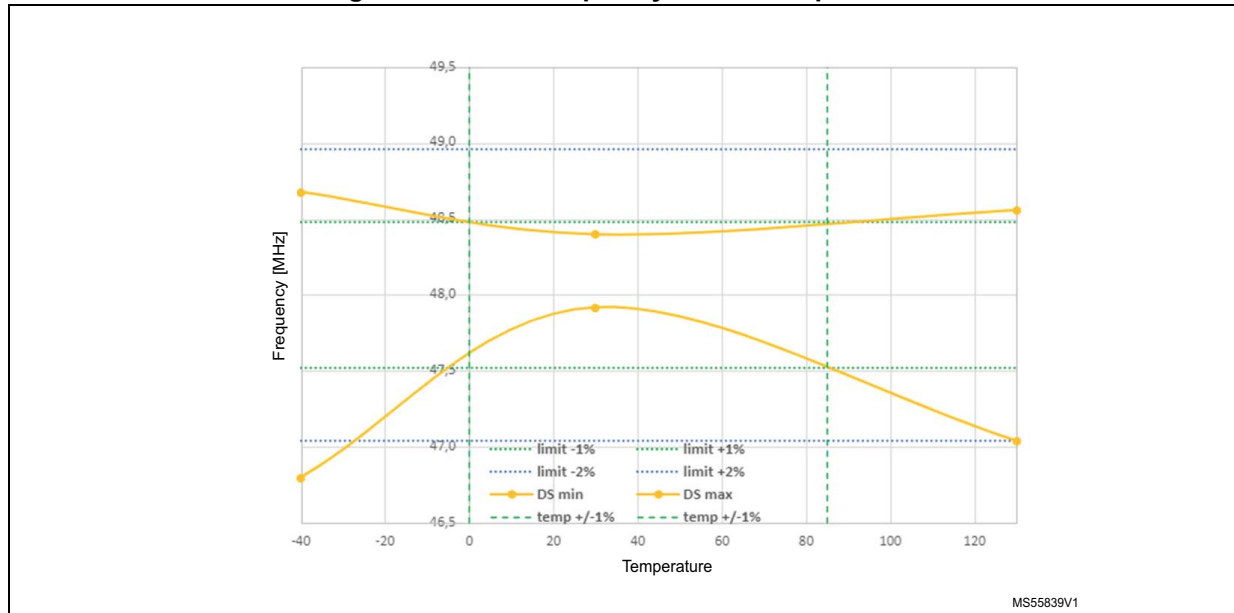


1. Based on characterization results, not tested in production
2. Specified by design. Not tested in production.

Figure 21. HSI48 frequency versus temperature



High-speed internal USB (HSIUSB48) oscillator

Table 42. HSIUSB48 oscillator characteristics

Symbol	Parameter	Conditions ⁽¹⁾	Min	Typ	Max	Unit
f_{HSIUSB48}	HSIUSB48 frequency	$V_{\text{DD}} = 3.0 \text{ V}$, $T_{\text{A}} = 30 \text{ °C}$	-	48	-	MHz
TRIM	HSIUSB48 user trimming step	-	-	0.11 ⁽²⁾	0.18 ⁽²⁾	%
USER TRIM COVERAGE	HSIUSB48 user trimming coverage	± 64 steps	± 6 ⁽³⁾	± 7 ⁽³⁾	-	%
$\text{DuCy}_{(\text{HSIUSB48})}$	Duty cycle	-	45 ⁽²⁾	-	55 ⁽²⁾	%
$\text{ACC}_{\text{HSIUSB48_REL}}$	Accuracy of the HSIUSB48 oscillator over temperature (factory calibrated)	$V_{\text{DD}} = 3.0 \text{ V to } 3.6 \text{ V}$ $T_{\text{A}} = -15 \text{ to } 85 \text{ °C}$	-	-	± 3	%
		$V_{\text{DD}} = 2.0 \text{ V to } 3.6 \text{ V}$ $T_{\text{A}} = -40 \text{ to } 125 \text{ °C}$	-	-	± 4.5 ⁽³⁾	
$\text{DV}_{\text{DD}(\text{HSIUSB48})}$	HSIUSB48 oscillator frequency drift with V_{DD}	$V_{\text{DD}} = 3 \text{ V to } 3.6 \text{ V}$	-	0.025 ⁽³⁾	0.05 ⁽³⁾	%
		$V_{\text{DD}} = 2.0 \text{ V to } 3.6 \text{ V}$	-	0.05 ⁽³⁾	0.1 ⁽³⁾	
$t_{\text{su}(\text{HSIUSB48})}$	HSIUSB48 oscillator start-up time	-	-	2.5 ⁽²⁾	6 ⁽²⁾	μs
$I_{\text{DD}(\text{HSIUSB48})}$	HSIUSB48 oscillator power consumption	-	-	340 ⁽²⁾	380 ⁽²⁾	μA